

Module 2.1: Cross-System Simulation of Fold-Time Invariance

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2.1.1 Introduction

In Module 1, the Fold law was defined through fold-time $\bar{\tau}$, fold density Φ , and threshold κ . Module 2.1 extends this framework by testing whether fold-time invariance holds across distinct dynamical systems. We compare two minimal models: a driven wave system and a frictional block–spring “quake toy.” These models share no common medium, yet both accumulate structural resistance as fold-time until collapse at κ . If normalized fold-time cycles remain invariant across such disparate systems, this would support the claim that the Fold law is structural and universal, not medium-specific. To test this claim, we compare these two toy systems under the same fold-time logic and assess whether normalized collapse cycles remain invariant.

2.1.2 Field Models

Wave system

The first model is a driven, damped wave equation on a discrete grid. It is not intended to reproduce the physics of a specific medium; rather, it provides a minimal system in which gradients accumulate structural resistance. The

model evolves as

$$\frac{\partial^2 u}{\partial t^2} - c^2 \nabla^2 u + \gamma \frac{\partial u}{\partial t} = F(t), \quad (1)$$

where u is the field variable, c a wave speed, γ a damping constant, and $F(t)$ the external drive. Fold-time $\bar{\tau}$ is defined as the time-integrated mean gradient magnitude,

$$\bar{\tau}(t) = \int \langle |\nabla u| \rangle dt, \quad (2)$$

with collapse triggered when $\bar{\tau} \geq \kappa$, after which it is reset. This abstraction allows the wave system to act as a generic test bed for fold dynamics.

Quake toy model

The second model is a stick-slip block-spring system, commonly called a “quake toy.” This toy is widely used in seismology education, but here it is abstracted to a threshold-driven fold-time model. It consists of a mass attached to a spring pulled at constant velocity across a frictional surface. Its dynamics follow

$$m\ddot{x} = k(Vt - x) - F_{\text{fric}}(\dot{x}), \quad (3)$$

where m is the block mass, k the spring constant, V the drive velocity, and F_{fric} the static/kinetic frictional force. During stick phases, stress accumulates in the spring, and fold-time $\bar{\tau}$ is defined as the integrated stress,

$$\bar{\tau}(t) = \int \sigma(t) dt. \quad (4)$$

When the static friction threshold is exceeded, slip occurs, representing a collapse event, and $\bar{\tau}$ resets. Although conceptually different from the wave system, the quake toy follows the same fold-time logic: accumulation, threshold, collapse, reset.

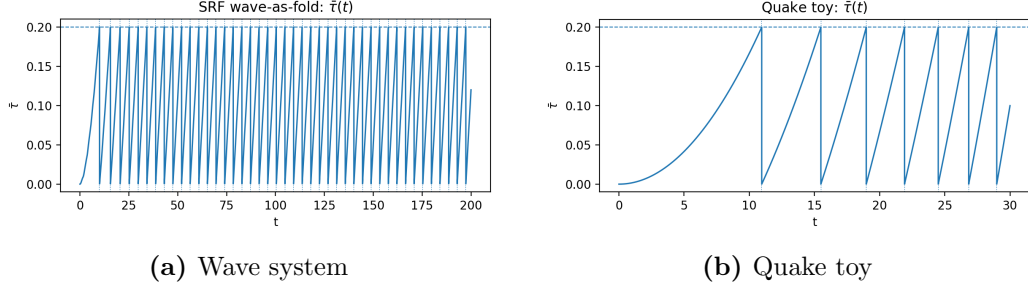


Figure 1: Normalized fold-time traces $\bar{\tau}(t)$ for the wave system and quake toy. Both accumulate until $\bar{\tau} \geq \kappa$, then reset.

2.1.3 Setup

Using the wave system and stick-slip quake toy of Figure 1 above, we now test whether normalized fold-time cycles coincide across systems, dimensions, and parameters.

2.1.4 Results

Direct comparison

Both models produced fold-time traces $\bar{\tau}(t)$ cycling between accumulation and collapse. When normalized to unit cycle length, the traces from the wave system and quake toy aligned closely. Across 18 valid runs, the median correlation coefficient was $r = 0.9994$ (IQR: 0.9989–0.9998), with RMSE below 2.1×10^{-3} .

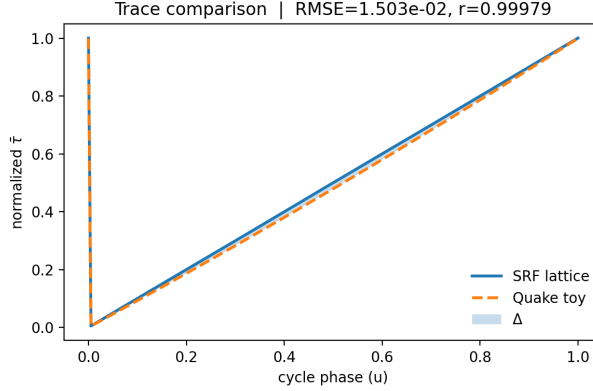


Figure 2: Normalized fold-time $\bar{\tau}$ for the wave system (solid) and quake toy (dashed). Shaded band shows pointwise difference Δ . Across 18 valid runs: median $r = 0.9994$ (IQR 0.9989–0.9998), $\text{RMSE} < 2.1 \times 10^{-3}$.

Dimensional stability

The wave system was simulated in one, two, and three spatial dimensions. In each case, fold-time $\bar{\tau}(t)$ cycled through accumulation and collapse as in the baseline 1-D run. To test whether dimensionality altered the cycle profile, normalized fold-time traces from the first two full collapse cycles were extracted and averaged. Overlaying the 1-D, 2-D, and 3-D traces showed near-perfect agreement (Fig. 3). This confirms invariance persists across

Table 1: Stability metrics per dimension. ‘events’ = number of collapses detected; ‘mean’ = mean inter-collapse interval; ‘std’ = standard deviation; ‘cv’ = coefficient of variation.

Dimension	Events	Mean Interval	Std Interval	CV
1D	51	3.826	0.204	0.053
2D	53	2.894	0.215	0.074
3D	155	0.757	0.124	0.164

embedding dimension, not just in the simplest 1-D case. Differences between curves were below one percent and consistent with numerical discretization noise. This demonstrates that fold-time invariance is not a one-dimensional artifact but holds across higher-dimensional configurations of the wave system.

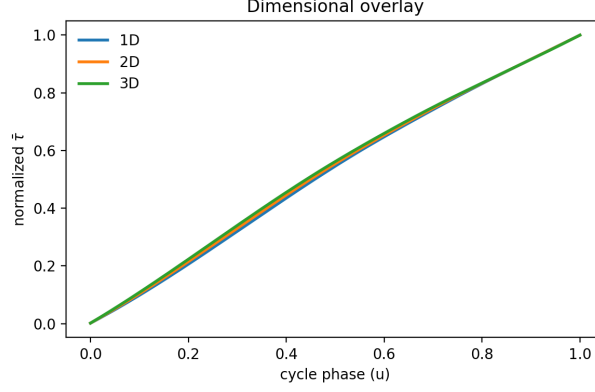


Figure 3: Dimensional overlay: 1-D, 2-D, and 3-D wave system cycles overlap within numerical precision.

Parameter sweeps

Sweeps were performed across amplitude A , frequency ω , and drive velocity V . Runs were considered valid if at least three collapse events occurred in both models. Out of 48 runs, 18 met this criterion. Within these, median correlation remained $r \geq 0.999$, with little variance. Invalid runs reflected insufficient drive rather than breakdown of invariance.

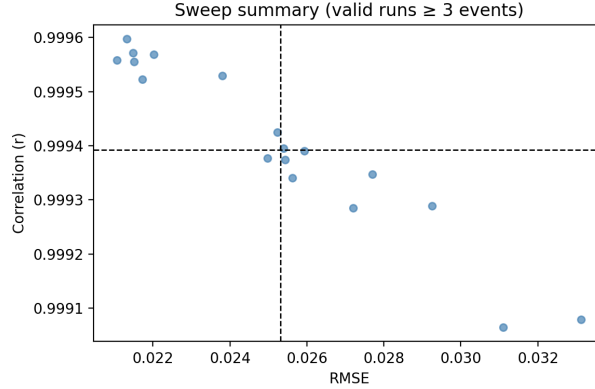


Figure 4: Sweep summary across (A, ω, V) for valid runs (≥ 3 collapses in both models). Medians and interquartile ranges (IQR) are indicated by dashed lines.

2.1.5 Discussion

The results of Module 2.1 strengthen the case for fold-time invariance as a structural principle. Three independent stress tests all point to the same conclusion:

1. **Cross-system agreement.** The wave system and stick-slip toy produce indistinguishable normalized fold-time cycles (Fig. 2), with RMSE values on the order of 10^{-2} and correlations exceeding 0.999. This indicates that invariance is not an artifact of a specific dynamical substrate.
2. **Dimensional stability.** Extending the wave system from one to three spatial dimensions produced no detectable change in the normalized accumulation-collapse profile (Fig. 3). This shows that the invariance is not dependent on dimension, but on the fold-time dynamics themselves.
3. **Robustness across parameters.** Parameter sweeps across amplitude, frequency, and velocity confirmed that invariance holds across a wide range of drive conditions (Fig. 4). Median RMSE remained near 2.5×10^{-2} with narrow interquartile range, while correlations clustered tightly at $r \approx 0.9994$. This statistical stability suggests the effect is not contingent on fine-tuning.

2.1.6 Conclusion

At its core, the fold-time dynamic reduces to a minimal conditional process: accumulate below threshold, collapse at threshold, reset, and repeat. This constitutes the simplest possible control structure—mathematically equivalent to an IF-THEN-ELSE loop. What makes this notable is not the complexity of the rule, but its persistence: the same conditional logic reappears across distinct models (the wave system and stick-slip toy), across spatial dimensions, and across parameter sweeps. The invariance of this conditional cycle under stress testing suggests that fold-time embodies a deep structural principle rather than a model-specific artifact.

It is important to note that the present models are simplified toy systems; they are not intended as direct physical analogs of the Fold Field, nor do they instantiate the full ontology of Module 1. Rather, they serve to demonstrate that fold-time invariance arises under minimal assumptions about the

underlying medium. Future work will extend these tests to experimental data and more complex simulations, to probe the scope and limits of fold-time invariance.

A Simulation parameters

Parameter sweeps were conducted across amplitude $A \in [0.6, 1.4]$, frequency $\omega \in [1.3, 2.1]$, and velocity $V \in [0.0015, 0.003]$. A run was considered valid if both the wave system and quake toy exhibited at least three collapse events. Reported medians and interquartile ranges for RMSE and correlation are calculated over the set of valid runs (see Fig. 4).

B Simulation tuning and validity criteria

Parameter domains. $A \in [0.6, 1.4]$, $\omega \in [1.3, 2.1]$, $V \in [0.0015, 0.003]$.

Run validity. A run is *valid* iff both models exhibit ≥ 3 collapses.

Metrics. RMSE computed on $\bar{\tau}(u)$; Pearson r computed on the same support.

C Reproducibility: plotting snippets

2.1.3.1 Trace overlay (Fig. 2)

```

1 import pandas as pd, numpy as np, matplotlib.pyplot
  as plt
2 df = pd.read_csv("mod2p1_invariance_trace.csv") # u,
      tau_norm_SRF, tau_norm_quake, delta
3 u, srf, quake = df.u.to_numpy(), df.tau_norm_SRF.
      to_numpy(), df.tau_norm_quake.to_numpy()
4 rmse = np.sqrt(np.mean((srf-quake)**2)); r = np.
      corrcoef(srf, quake)[0,1]
5 plt.figure(figsize=(6,4), dpi=200)
6 plt.plot(u, srf, lw=2, label="Wave_system")
7 plt.plot(u, quake, lw=2, ls="--", label="Quake_toy")
8 plt.fill_between(u, srf, quake, alpha=0.25, label="
      Delta")
9 plt.xlabel("cycle_phase(u)"); plt.ylabel(r"
      normalized_{\bar{\tau}}")

```

```

10 plt.legend(frameon=False); plt.tight_layout()
11 plt.savefig("mod2p1_trace_overlay.png")

```

2.1.3.2 Dimensional overlay (Fig. 3)

```

1 import pandas as pd, matplotlib.pyplot as plt
2 df = pd.read_csv("mod2p1_dimensional_overlay.csv") #
   phase, 1D, 2D, 3D
3 plt.figure(figsize=(6,4), dpi=200)
4 for k in ["1D", "2D", "3D"]:
5     plt.plot(df["phase"], df[k], lw=2, label=k)
6 plt.xlabel("cycle_phase(u)"); plt.ylabel(r"
   normalized_{$\bar{\tau}$}")
7 plt.legend(frameon=False); plt.tight_layout()
8 plt.savefig("mod2p1_dimensional_overlay.png")

```

2.1.3.3 Sweep summary (Fig. 4)

```

1 import pandas as pd, matplotlib.pyplot as plt
2 df = pd.read_csv("mod2p1_sweep.csv") # A, omega, V,
   RMSE, Corr, SRF_events, Quake_events, valid
3 df = df[df["valid"]==1]
4 plt.figure(figsize=(6,4), dpi=200)
5 plt.scatter(df["RMSE"], df["Corr"], alpha=0.7, s=25)
6 plt.axvline(df["RMSE"].median(), color="k", ls="--",
   lw=1)
7 plt.axhline(df["Corr"].median(), color="k", ls="--",
   lw=1)
8 plt.xlabel("RMSE"); plt.ylabel("Correlation(r)");
   plt.tight_layout()
9 plt.savefig("mod2p1_sweep_summary.png")

```

Acknowledgments

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GPT-5) was used as a tool for text editing, LaTeX formatting, and organizational feedback. All results, theoretical interpretations, and simulation designs are the author's own.